

Django Templates: Advanced Tags & Filters

Ye document Django templates ke advanced concepts aur rare tags par focus karta hai. Har topic ke saath visual structure aur step-by-step dry run diya gaya hai taaki concepts easily samajh aa jayein.

1. Regroup Tag

Kya hai? Ye tag ek list of dictionaries (ya objects) ko kisi specific attribute (jaise 'class', 'category') ke basis par group karne ke kaam aata hai.

Step 1 (View mein data create karna):

```
students_list = [ {'name': 'Mohit', 'class': '10th'}, {'name': 'Rohit', 'class': '9th'}, {'name': 'Sohit', 'class': '8th'} ]
```

Step 2 (Template mein Tag use karna):

```
{% regroup students_list by class as grouped_students %}
```

Logic: Ab `grouped\_students` variable ke andar data unki class ke hisaab se group ho chuka hai.

Step 3 (Nested For Loop lagana):

```
<ul> {% for group in grouped_students %} <li>Class: {{ group.grouper }}
<ul> {% for student in group.list %} <li>{{ student.name }}</li> {% endfor %}
</ul> </li> {% endfor %} </ul>
```

Result: Pehle "Class 10th" print hoga, fir uske andar ke students (jaise Mohit). Yahan `group.grouper` wo attribute hai jisme group kiya gaya tha ('class').

2. Widthratio Tag

Kya hai? Ye ek math helper tag hai jo simple percentage calculation ke kaam aata hai (jaise HTML progress bar ki width set karne ke liye).

Formula: $(\text{first_value} / \text{second_value}) * \text{max_width}$

3. Spaceless Tag

Kya hai? Ye tag HTML tags ke beech ke extra white spaces ya line breaks ko remove kar deta hai, jisse browser mein output cleanly render hota hai.

Before Spaceless (Code block):

```
{% spaceless %} <div> <span> This is spaceless example </span> </div> {% endspaceless %}
```

After Render: HTML structure ek line mein aa jayega -> `<div><span> This is spaceless example </span></div>`. Dhyan rahe, text ke andar ka space remove nahi hota, sirf HTML tags ke beech ka space hataaya jata hai.

4. Filters: Block vs Inline

Filters data ko render hone se pehle modify ya format karne ke kaam aate hain.

A. Block Filter

Jab aapko multiple lines of text par ek sath filter lagana ho:

```
{% filter upper %} This text is converted to uppercase. {% endfilter %}
```

Output: THIS TEXT IS CONVERTED TO UPPERCASE.

B. Inline Filter

Jab aapko directly variable render karte time filter lagana ho (pipe | symbol use karke):

```
{{ student.name|lower }}
```

Output: Agar name "Mohit" tha, toh output "mohit" aayega.

Extra parameters: Addition karne ke liye -> `{{ class|add:"1" }}`

5. Dynamic CSS in Templates

Static files (jaise CSS) ko base template aur child templates mein alag-alag dynamically kaise use karein?